djgrab — Web Download Implementation Plan

Personal, non-commercial tool · adds yt-dlp–based web downloads (YouTube / SoundCloud) to the funnel: recognize → stream → own only keepers · date: 2026-10-15

Decisions

- **Sources:** YouTube + SoundCloud, via `yt-dlp` as an external binary.
- **Format:** MP3/AAC (`m4a` default), tagged `quality=lossy` — no fake-lossless files.
- **Trigger:** auto-download for keepers (between "free" and "buy list") + manual `djgrab grab <id>`.
- **Sets:** a set link yields every identified segment as its own track file, like a playlist. Cutting ships with M4 (needs the fusion timeline).

0. Doc edits first (spec stays source of truth)

docs/SPEC.md

- §1: new outcome — kept tracks and every identified set segment are downloaded from the web as lossy copies when no free source exists; buy links remain the lossless upgrade path.
- §2 table: set input row → tracklist extraction → N track jobs + set audio download → N per-track files.
- §3 diagram: add branch `keeper? → free links → web download → staging → inbox`.
- §4: `tracks` gains `acquired_via TEXT DEFAULT 'inbox'` (`inbox|free|web|purchase`).
- §8: new step 3b — Web download: yt-dlp search → metadata-only candidates → §7 match rules → download → staging → inbox. Failure/ambiguity → `needs_review` or fall back to buy list.
- §9: new post-fusion step — Set audio + cutting: yt-dlp downloads the set audio once; ffmpeg cuts one file per identified segment (`trim_s` inward); watchlist ID clips cut from the same audio.
- §12 provider table: row `yt-dlp | none | ToS-gray, IP throttling | metadata-first, daily cap, subprocess retries`.
- §13: new config block (below). §14: add `acquire/webdl.py`, `media/cut.py`.
- §15: insert M3.5, amend M4 acceptance, note M5 shrink (watchlist clips come free from set audio).

CLAUDE.md

- Allowed audio inputs: add files the tool downloads itself via yt-dlp (single-track rips and set audio for cutting); personal use; always flagged lossy.
- Stack → External binaries: add `yt-dlp`. Commands: add `uv run djgrab grab <track_id|job_id>`.

- **docs/DECISIONS.md** entries (date, reason): ToS acceptance for personal format-shifting; m4a over FLAC for rips; `auto_keepers` trigger policy; `trim_s=2` rationale.

1. Config (§13)

```
download:
  enabled: true
  sources: [youtube, soundcloud]
  format: m4a              # m4a | mp3
  bitrate: 256k
  max_per_day: 20          # enforced via existing `spend` table, provider='ytdlp'
  trim_s: 2                # seconds trimmed inside each set-segment boundary
  singles: auto_keepers    # auto_keepers | on_demand
  fallback_to_buylist: true
```

`download.enabled: false` disables the whole feature with one switch.

2. New modules

acquire/webdl.py — interface + yt-dlp subprocess impl:

```
class Downloader(Protocol):
    def search_meta(self, query: str, limit: int) -> list[WebCandidate]: ...
    def fetch_track(self, url: str, dest_dir: Path) -> Path: ...  # bestaudio → m4a/mp3
    def fetch_set_audio(self, url: str, dest: Path) -> Path: ...  # bestaudio, keep con

@dataclass
class WebCandidate: source: str; url: str; title: str; uploader: str; duration_s: int
```

- `YtDlpDownloader` shells out: `yt-dlp "ytsearch5:{q}" --dump-json` (and `scsearch5:`), one candidate per JSON line; `fetch_track` = `-f bestaudio -x --audio-format {format} --audio-quality {bitrate}`.
- Subprocess runner: timeout, 1 concurrent launch, retry ×3 with backoff; failures → job `failed` (retryable), per CLAUDE.md error rule.

media/cut.py — set slicing:

- Input: set audio path + `[(start_s, end_s, track_id)]` from fusion output.
- Per segment: `ffmpeg -ss {start+trim} -to {end-trim} -i set_audio -c:a aac -b:a {bitrate} out.m4a`; skip segments < 60 s after trim; each cut lands in staging with its `track_id`.

3. Flow wiring

Singles (M3.5): keeper promotion (or `djgrab grab`) → if no free link → `webdl.search_meta("{artist} {title} {mix}")` per source → candidates through existing `identity/match.py` rules

(token ≥90, mix equality, duration ±2s/±3s) + normalize.py title parsing → best match downloads to `paths.data/staging/` → insert `inbox_files` row with `matched_track_id` preset, `status=ready` → existing inbox pipeline converts/tags/files it (quality check flags `lossy`) → `acquired_via='web'`, job `acquired`. No/ambiguous match → `needs_review` (candidates shown) or, on hard failure, back to buy list per `fallback_to_buylist`.

Inbox change (one branch): a row with preset `matched_track_id` skips the match phase entirely.

Sets (with M4): fusion timeline complete → `fetch_set_audio` → `media/cut.py` → every cut staged with its `track_id` → same inbox path. ID segments: 12 s clips cut from the same audio (feeds §9.6 watchlist at zero provider cost).

Migration (post-M1, user-approved): `ALTER TABLE tracks ADD COLUMN acquired_via ...`.

4. CLI + doctor

- `djgrab grab <track_id|job_id>`: on-demand download regardless of keeper status; counts toward `max_per_day`, clear error pointing at config when capped.
- `djgrab doctor`: checks `yt-dlp` presence/version alongside ffmpeg/flac/fpcalc (not currently installed on this machine — output includes install guidance).

5. Build order

1. Doc edits (SPEC, CLAUDE, DECISIONS).
2. Config block + migration (`acquired_via`).
3. `Downloader` interface + `YtDlpDownloader` + doctor check.
4. Search→match→staging→inbox wiring + `needs_review`/buylist fallback.
5. `grab` command + daily cap via `spend`.
6. M3.5 tests + acceptance.
7. *After M4 fusion exists:* `media/cut.py` + set-audio step + golden test.

6. Tests (no live network, per CLAUDE.md)

- Fixture: recorded `--dump-json` search output for invented §16 tracks ("Nova Ridge — Solar Drift"); match-unit tests reuse §7 patterns (right version, wrong version, ambiguous → review).
- `Downloader` mocked at interface level (subprocess, so not `respx`).
- Golden set test: concatenate generated tones (alternating 440/550 Hz, known boundaries) into a fake set WAV → `cut.py` → assert file count, durations ±tolerance, dominant frequency per cut (numpy already in stack).
- Integration: staged file with preset `matched_track_id` files correctly, `acquired_via='web'`, `DJGRAB_QUALITY=lossy` (reuse M1 low-pass fixture).

7. Acceptance

M3.5 — Web download (singles): mocked tests green; one real E2E (manual, network): keeper → filed m4a, correct tags, `quality=lossy`; ambiguous search → `needs_review` with candidates; cap blocks download #21 of the day with a clear message; missing yt-dlp → doctor error + graceful job failure, batch survives.

M4 amendment — set cutting: offline golden test passes; real-set E2E (manual): set link → N identified tracks each filed as separate lossy m4a, set playlist + XML updated, ID clips saved for watchlist.

Risks

- YouTube downloads violate YT ToS — accepted for personal format-shifting; recorded in DECISIONS.md.
- yt-dlp breaks often (extractor churn, throttling) — routine updates; failures degrade to `needs_review`/buylist, never crash the batch.
- Set cuts always carry a few seconds of transition bleed — inherent to cutting a continuous mix; `trim_s` mitigates, tune on real sets.